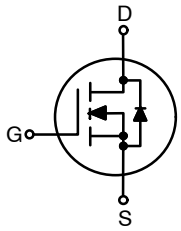
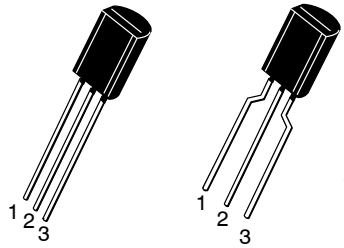
N-Channel Enhancement Mode Field Effect Transistor

2N7000, 2N7002, NDS7002A





1 – Source
2 – Gate
3 – Drain

TO–92
CASE 135AN

TO–92
CASE 135AR

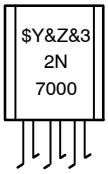
Description

These N–channel enhancement mode field effect transistors are produced using **onsemi**'s proprietary, high cell density, DMOS technology. These products have been designed to minimize on–state resistance while providing rugged, reliable, and fast switching performance. These products are particularly suited for low–voltage, low–current applications, such as small servo motor control, power MOSFET gate drivers, and other switching applications.

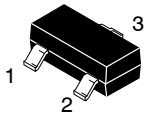
Features

- High Density Cell Design for Low $R_{DS(on)}$
- Voltage Controlled Small Signal Switch
- Rugged and Reliable
- High Saturation Current Capability
- ESD Protection Level: HBM > 100 V, CDM > 2 kV
- This Device is Pb–Free and Halogen Free

MARKING DIAGRAM



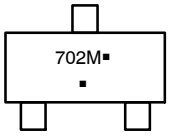
$Y = **onsemi** Logo
&Z = Assembly Plant Code
&3 = Date Code
2N7000 = Specific Device Code



1 – Gate
2 – Source
3 – Drain

SOT–23
CASE 318–08

MARKING DIAGRAM



702 = Specific Device Code
M = Date Code
▪ = Pb–Free Package

(Note: Microdot may be in either location)

ORDERING INFORMATION

See detailed ordering and shipping information on page 7 of this data sheet.

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ABSOLUTE MAXIMUM RATINGS Values are at T_C = 25°C unless otherwise noted.

Symbol	Parameter	Value			Unit
		2N7000	2N7002	NDS7002A	
V_{DSS}	Drain−to−Source Voltage	60			V
V_{DGR}	Drain−Gate Voltage (R_{GS} ≤ 1 MW)	60			V
V_{GSS}	Gate−Source Voltage − Continuous	±20			V
	Gate−Source Voltage − Non Repetitive (tp < 50 ms)	±40			
I_D	Maximum Drain Current − Continuous	200	115	280	mA
	Maximum Drain Current − Pulsed	500	800	1500	
P_D	Maximum Power Dissipation Derated above 25°C	400	200	300	mW
		3.2	1.6	2.4	mW/°C
T_J, T_{STG}	Operating and Storage Temperature Range	−55 to 150		−65 to 150	°C
T_L	Maximum Lead Temperature for Soldering Purposes, 1/16−inch from Case for 10 s	300			°C

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

THERMAL CHARACTERISTICS Values are at T_C = 25°C unless otherwise noted.

Symbol	Parameter	Value			Unit
		2N7000	2N7002	NDS7002A	
$R_{\theta JA}$	Thermal Resistance, Junction to Ambient	312.5	625	417	°C/W

ELECTRICAL CHARACTERISTICS
Values are at T_C = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Type	Min	Typ	Max	Unit
OFF CHARACTERISTICS							
BV_{DSS}	Drain−Source Breakdown Voltage	V_{GS} = 0 V, I_D = 10 μA	All	60	–	–	V
I_{DSS}	Zero Gate Voltage Drain Current	V_{DS} = 48 V, V_{GS} = 0 V	2N7000	–	–	1	μA
		V_{DS} = 48 V, V_{GS} = 0 V, T_C = 125°C		–	–	1	mA
		V_{DS} = 60 V, V_{GS} = 0 V	2N7002 NDS7002A	–	–	1	μA
		V_{DS} = 60 V, V_{GS} = 0 V, T_C = 125°C		–	–	0.5	mA
I_{GSSF}	Gate − Body Leakage, Forward	V_{GS} = 15 V, V_{DS} = 0 V	2N7000	–	–	10	nA
		V_{GS} = 20 V, V_{DS} = 0 V	2N7002 NDS7002A	–	–	100	
I_{GSSR}	Gate − Body Leakage, Reverse	V_{GS} = −15 V, V_{DS} = 0 V	2N7000	–	–	−10	nA
		V_{GS} = −20 V, V_{DS} = 0 V	2N7002 NDS7002A	–	–	−100	
ON CHARACTERISTICS							
$V_{GS(th)}$	Gate Threshold Voltage	V_{DS} = V_{GS}, I_D = 1 mA	2N7000	0.8	2.1	3	V
		V_{DS} = V_{GS}, I_D = 250 μA	2N7002 NDS7002A	1	2.1	2.5	

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ELECTRICAL CHARACTERISTICS (continued)
Values are at T_C = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Type	Min	Typ	Max	Unit
ON CHARACTERISTICS							
$R_{DS(on)}$	Static Drain−Source On−Resistance	V_{GS} = 10 V, I_D = 500 mA	2N7000	–	1.2	5	Ω
		V_{GS} = 10 V, I_D = 500 mA, T_C = 125°C		–	1.9	9	
		V_{GS} = 4.5 V, I_D = 75 mA		–	1.8	5.3	
		V_{GS} = 10 V, I_D = 500 mA	2N7002	–	1.2	7.5	
		V_{GS} = 10 V, I_D = 500 mA, T_C = 100°C		–	1.7	13.5	
		V_{GS} = 5 V, I_D = 50 mA		–	1.7	7.5	
		V_{GS} = 5 V, I_D = 50 mA, T_C = 100°C		–	2.4	13.5	
		V_{GS} = 10 V, I_D = 500 mA	NDS7002A	–	1.2	2	
		V_{GS} = 10 V, I_D = 500 mA, T_C = 125°C		–	2	3.5	
		V_{GS} = 5 V, I_D = 50 mA		–	1.7	3	
		V_{GS} = 5 V, I_D = 50 mA, T_C = 125°C		–	2.8	5	
$V_{DS(on)}$	Drain−Source On−Voltage	V_{GS} = 10 V, I_D = 500 mA	2N7000	–	0.6	2.5	V
		V_{GS} = 4.5 V, I_D = 75 mA		–	0.14	0.4	
		V_{GS} = 10 V, I_D = 500 mA	2N7002	–	0.6	3.75	
		V_{GS} = 5.0 V, I_D = 50 mA		–	0.09	1.5	
		V_{GS} = 10 V, I_D = 500 mA	NDS7002A	–	0.6	1	
		V_{GS} = 5.0 V, I_D = 50 mA		–	0.09	0.15	
g_{FS}	Forward Transconductance	V_{DS} = 10 V, I_D = 200 mA	2N7000	100	320	–	mS
		$V_{DS} \geq 2\ V_{DS(on)}$, I_D = 200 mA	2N7002	80	320	–	
		$V_{DS} \geq 2\ V_{DS(on)}$, I_D = 200 mA	NDS7002A	80	320	–	
DYNAMIC CHARACTERISTICS							
C_{iss}	Input Capacitance	V_{DS} = 25 V, V_{GS} = 0 V, f = 1.0 MHz	All	–	20	50	pF
C_{oss}	Output Capacitance		All	–	11	25	
C_{rss}	Reverse Transfer Capacitance		All	–	4	5	
t_{on}	Turn−On Time	V_{DD} = 15 V, R_L = 25 Ω, I_D = 500 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	2N7000	–	–	10	ns
		V_{DD} = 30 V, R_L = 150 Ω, I_D = 200 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	2N7002 NDS7002A	–	–	20	
t_{off}	Turn−Off Time	V_{DD} = 15 V, R_L = 25 Ω, I_D = 500 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	2N7000	–	–	10	ns
		V_{DD} = 30 V, R_L = 150 Ω, I_D = 200 mA, V_{GS} = 10 V, R_{GEN} = 25 Ω	2N7002 NDS7002A	–	–	20	
DRAIN−SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS							
I_S	Maximum Continuous Drain−Source Diode Forward Current		2N7002	–	–	115	mA
			NDS7002A	–	–	280	

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ELECTRICAL CHARACTERISTICS (continued)
Values are at T_C = 25°C unless otherwise noted.

Symbol	Parameter	Conditions	Type	Min	Typ	Max	Unit
DRAIN−SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS							
I_{SM}	Maximum Pulsed Drain−Source Diode Forward Current		2N7002	–	–	0.8	A
			NDS7002A	–	–	1.5	
V_{SD}	Drain−Source Diode Forward Voltage	V_{GS} = 0 V, I_S = 115 mA (Note 1)	2N7002	–	0.88	1.5	V
		V_{GS} = 0 V, I_S = 400 mA (Note 1)	NDS7002A	–	0.88	1.2	

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

1. Pulse test: Pulse Width ≤ 300 μs, Duty Cycel ≤ 2 %

TYPICAL PERFORMANCE CHARACTERISTICS

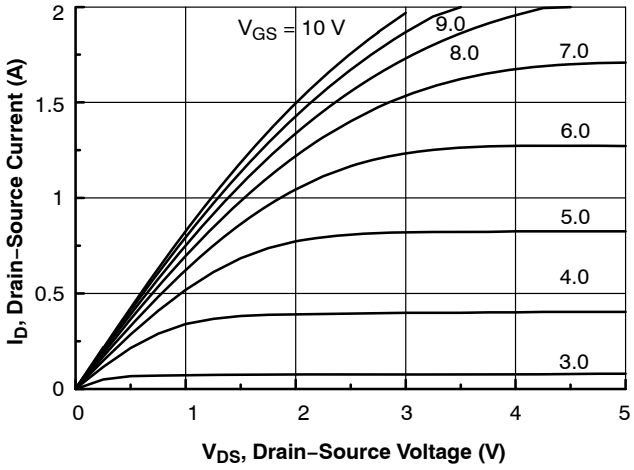

Figure 1. On−Region Characteristics

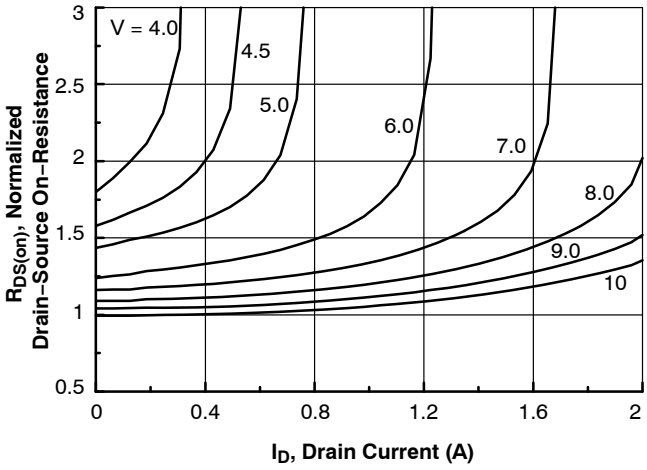

Figure 2. On−Resistance Variation with Gate Voltage and Drain Current

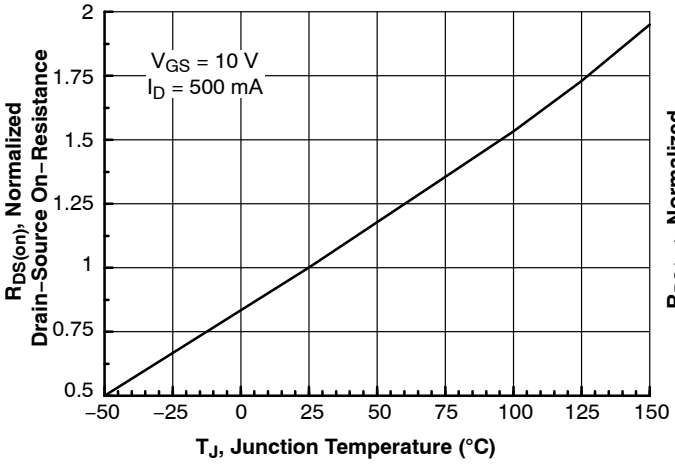

Figure 3. On−Resistance Variation with Temperature

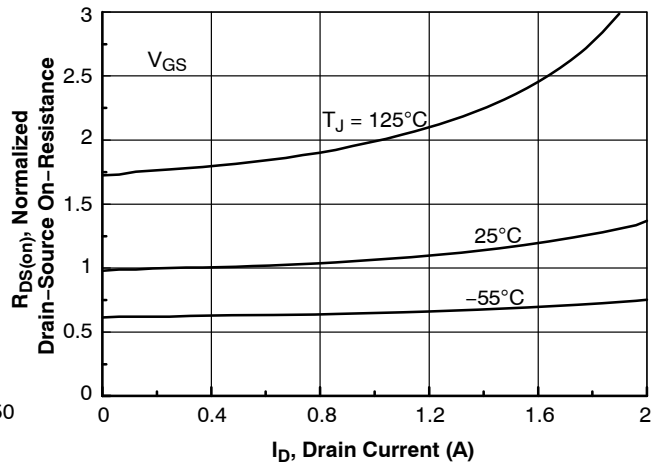

Figure 4. On−Resistance Variation with Drain Current and Temperature

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TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

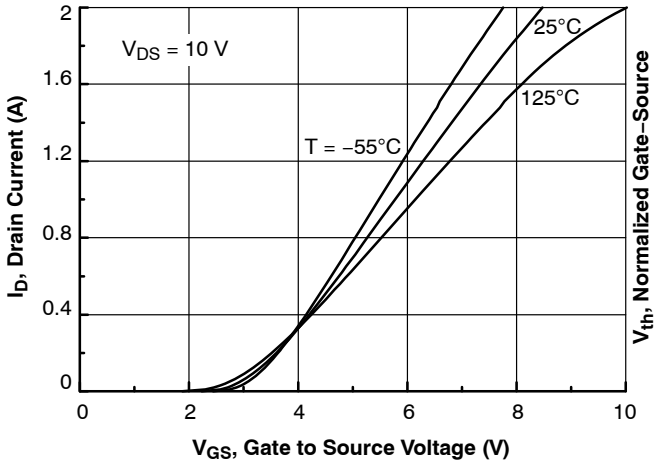

Figure 5. Transfer Characteristics

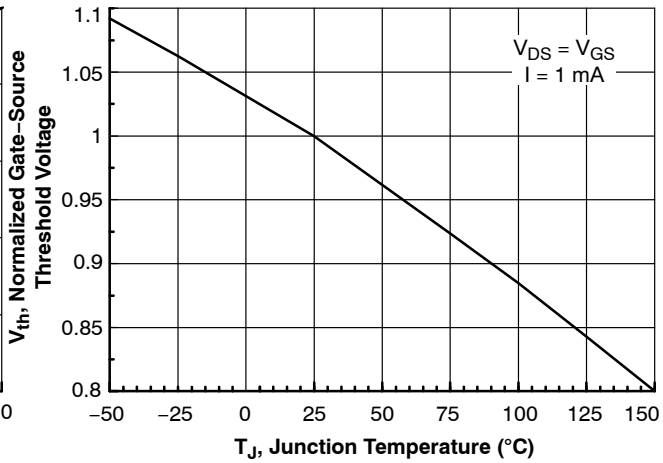

Figure 6. Gate Threshold Variation with Temperature

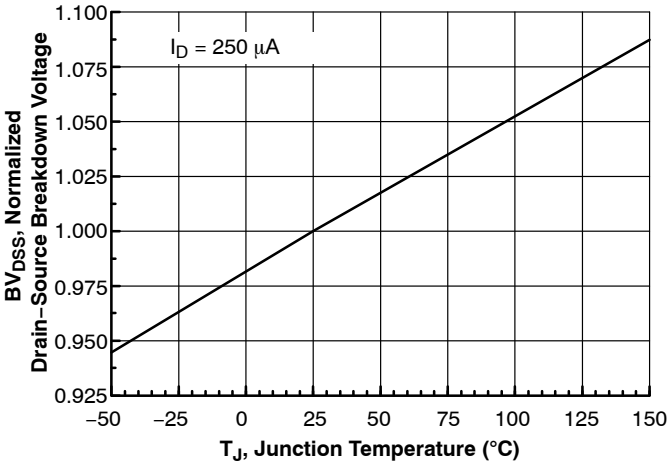

Figure 7. Breakdown Voltage Variation with Temperature

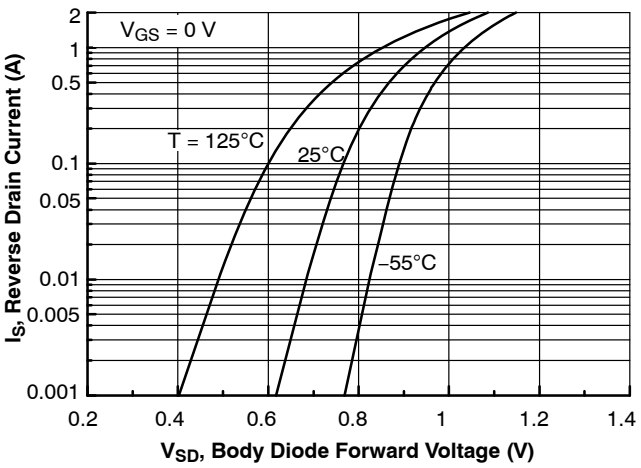

Figure 8. Body Diode Forward Voltage Variation with

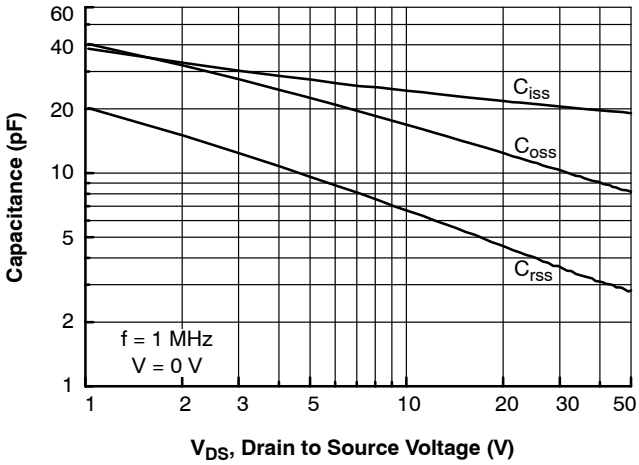

Figure 9. Capacitance Characteristics

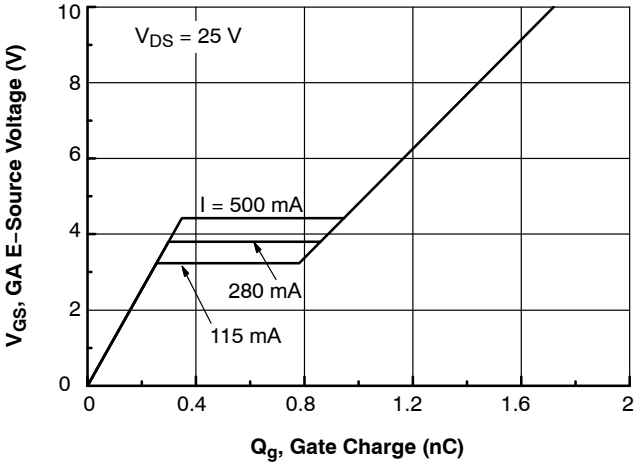

Figure 10. Gate Charge Characteristics

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TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

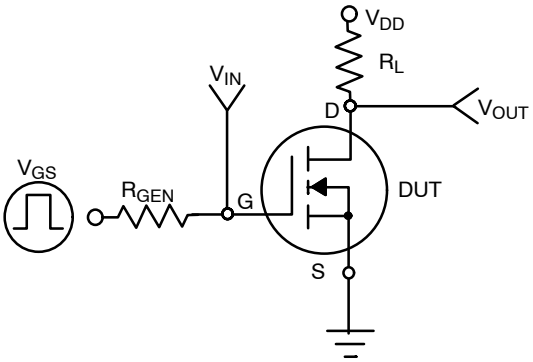

Figure 11. Switching Test Circuit

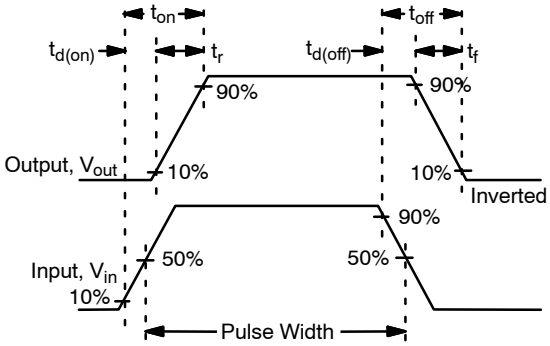

Figure 12. Switching Waveforms

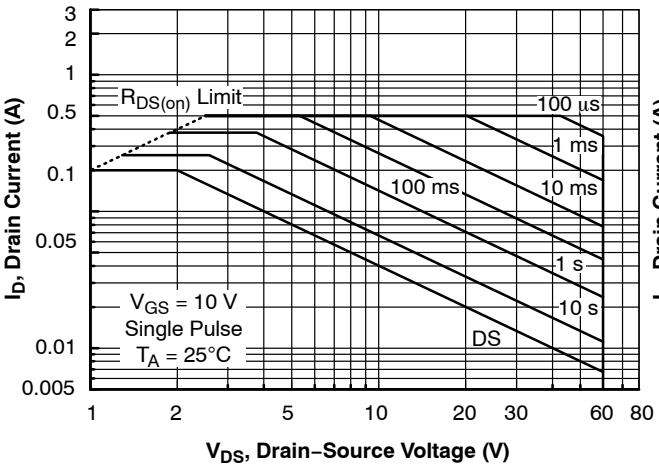

Figure 13. 2N7000 Maximum Safe Operating Area

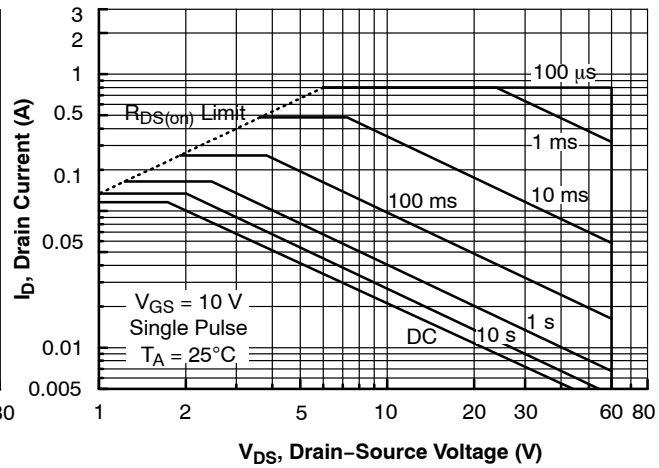

Figure 14. 2N7002 Maximum Safe Operating Area

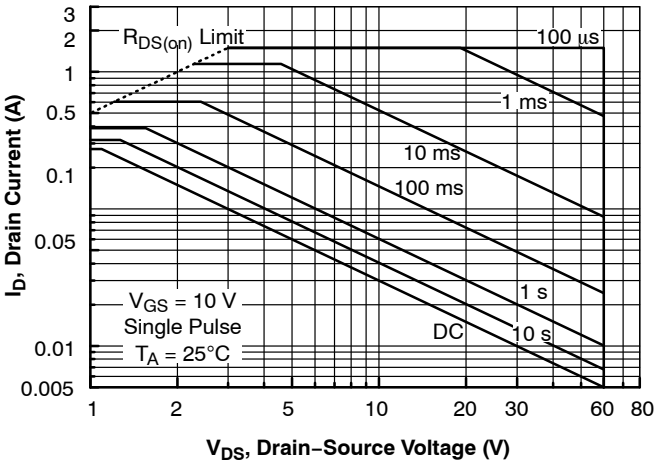

Figure 15. NDS7000A Maximum Safe Operating Area

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TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

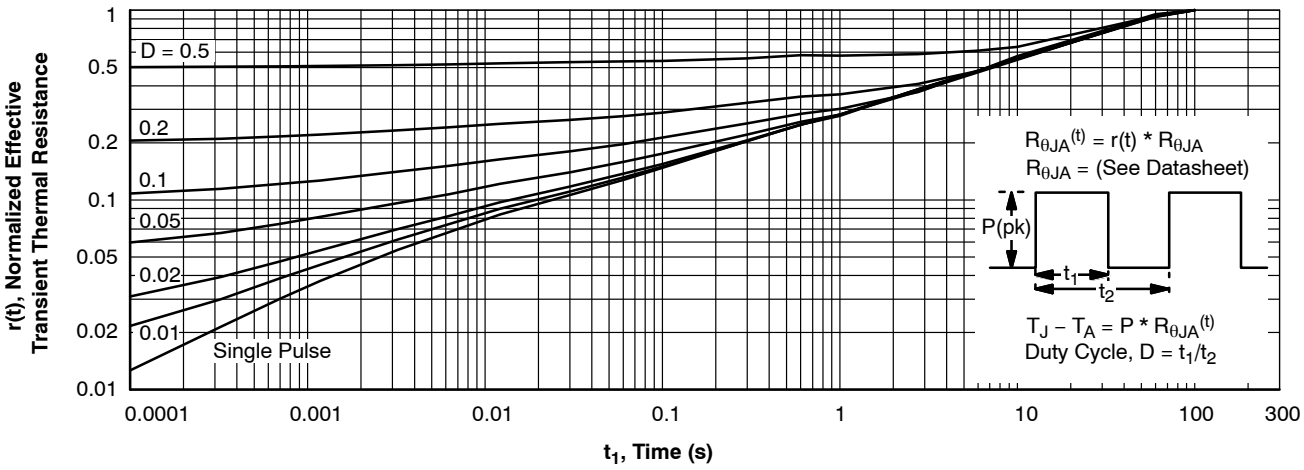

Figure 16. TO−92, 2N7000 Transient Thermal Response Curve

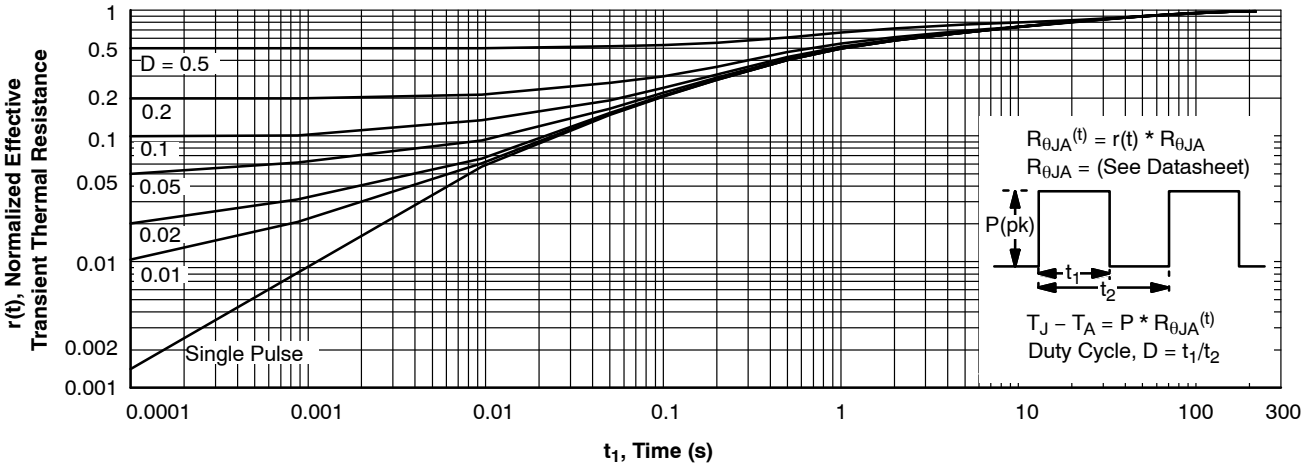

Figure 17. SOT−23, 2N7002 / NDS7002A Transient Thermal Response Curve

ORDERING INFORMATION

Part Number	Marking	Package	Packing Method†	Min Order Qty / Immediate Pack Qty
2N7000	2N7000	TO−92 3L (Pb−Free)	Bulk	10000 / 1000
2N7000−D74Z			Ammo	2000 / 2000
2N7000−D75Z			Tape and Reel	2000 / 2000
2N7000−D26Z				2000 / 2000
2N7002	702	SOT−23 3L (Pb−Free)	Tape and Reel	3000 / 3000
NDS7002A	712			3000 / 3000

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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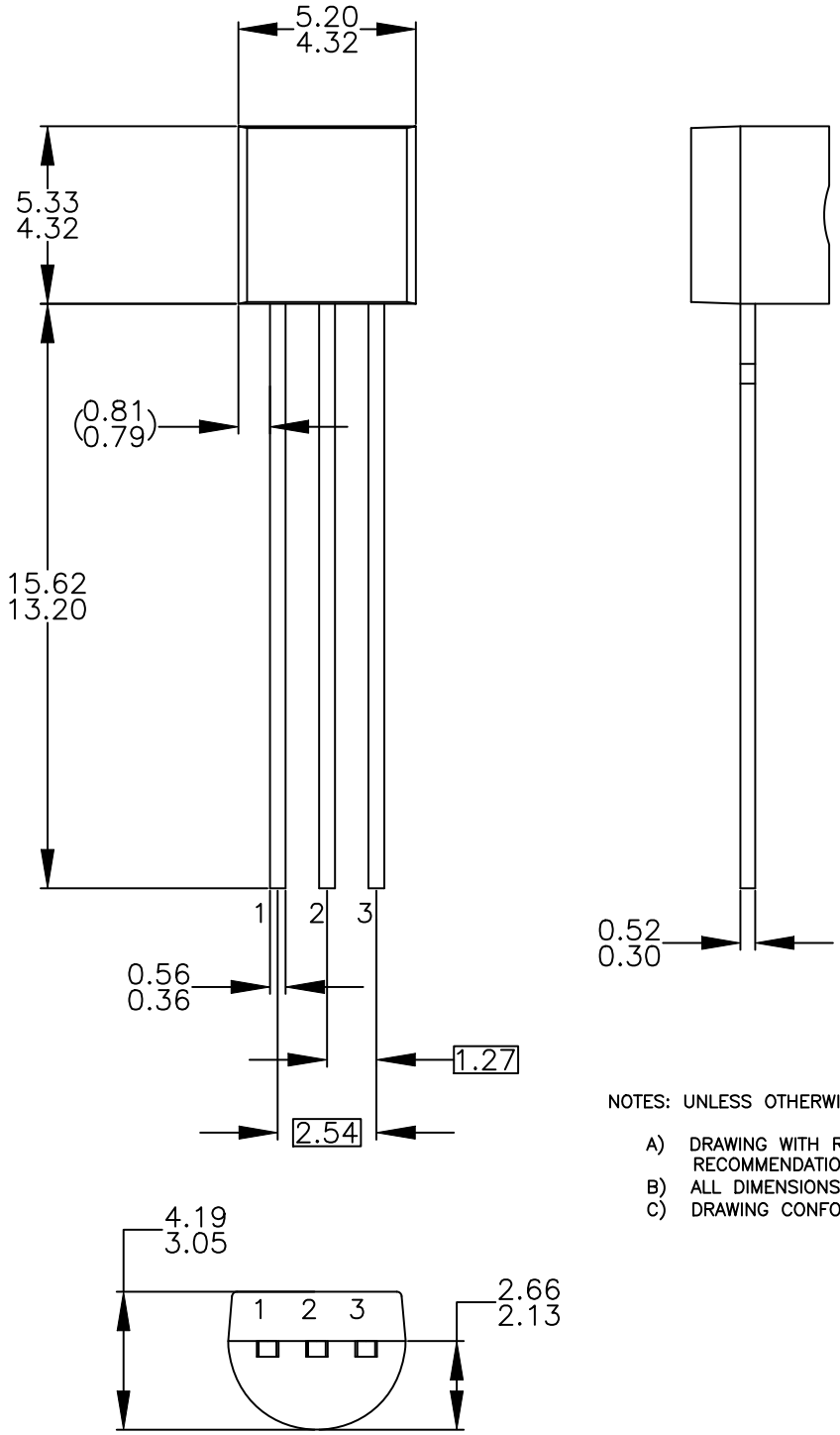
TO−92 3 4.825x4.76
CASE 135AN
ISSUE O

DATE 31 JUL 2016



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C) DRAWING CONFORMS TO ASME Y14.5M−2009.

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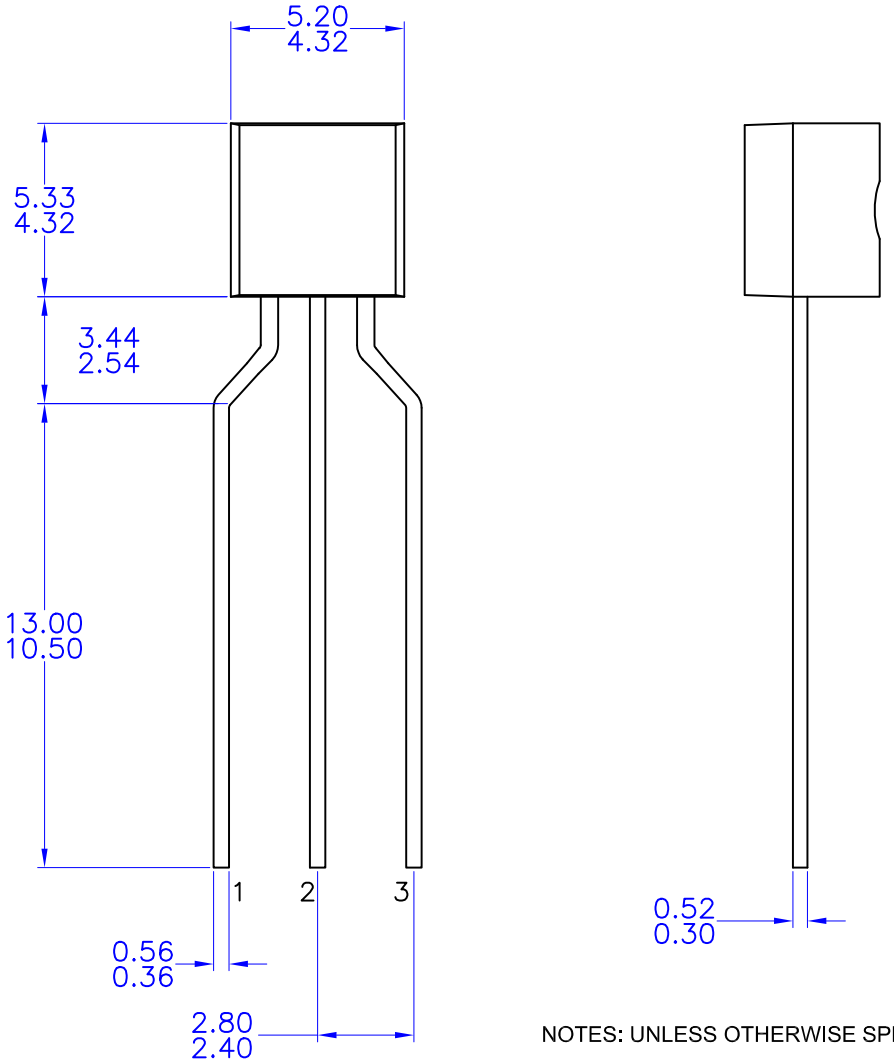
TO−92 3 4.83x4.76 LEADFORMED
CASE 135AR
ISSUE O

DATE 30 SEP 2016



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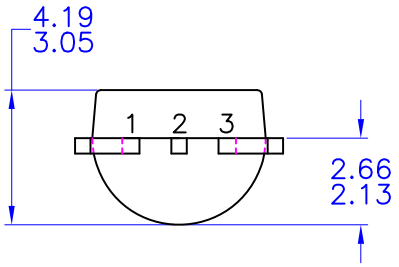

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